M. BASHEER ALI

STORAGE QA ENGINEER
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PROFESSIONAL PROFILE

Storage QA Engineer with around 3 years of experience validating enterprise storage arrays, SSD/HDD/NVMe drives, RAID, Fibre Channel, and iSCSI environments. Skilled in Python/Pytest automation, Linux debugging, FIO performance testing, and storage protocol validation. Experienced in developing automation frameworks that improve regression efficiency and ensure high-quality enterprise storage releases.

PROFESSIONAL SUMMARY

- Proficient in Python with strong knowledge of OOP concepts, data structures, file handling, exception handling, regular expressions, and scripting.
- Experienced in developing and maintaining Python/Pytest automation frameworks for storage validation, regression testing, log collection, and result analysis.
- Proficient in Python modules including os, sys, subprocess, argparse, hashlib, logging, datetime, json, re, and Paramiko for automation, remote execution, and log analysis.
- Hands-on experience in validating enterprise storage platforms, focusing on drive qualification, compatibility, firmware, reliability, performance, and fault recovery.
- Strong understanding of Functional, Integration, System, Regression, Performance, Stress, and Re-testing methodologies with knowledge of SDLC, STLC, Agile, and Defect Life Cycle.
- Experienced in designing, executing, and automating test cases for drive validation, RAID validation, firmware validation, and storage feature testing.
- Strong knowledge of enterprise storage technologies including HDD, SSD, NVMe, Storage Arrays, JBOD, DAS, SAN, and NAS.
- Familiar with storage protocols including SATA, SAS, SCSI, NVMe, iSCSI, and Fibre Channel (FC).
- Skilled in storage performance analysis using FIO, evaluating IOPS, throughput, latency, bandwidth, and queue-depth metrics.
- Experienced in storage diagnostics using smartctl, nvme-cli, storcli, lsblk, lspci, dmesg, and Linux system logs for troubleshooting and root-cause analysis.
- Good understanding of SSD technologies including wear leveling, TRIM, garbage collection, write amplification, over-provisioning, endurance, and SMART health monitoring.
- Strong knowledge of RAID technologies including RAID levels, Hot Spare, Online RAID Level Migration (ORLM), Capacity Expansion, Consistency Check, Rebuild, and Hot Swapping.
- Proficient with Linux, Shell Scripting, Git, Jenkins, JIRA, PyCharm, PuTTY, VMware, and x86 server firmware components including BIOS, BMC, and CPLD.
- Leveraged GitHub Copilot and Claude to accelerate automation development, log analysis, and debugging.

TECHNICAL SKILLS

Storage Management: DAS, SAN, Storage Arrays
Storage Technologies: HDD, SSD, SATA, SAS, iSCSI, RAID, NVMe
Storage & Testing Tools: IOmeter, FIO, mdadm, storcli, smartctl, IPMI, JIRA
Programming & Automation: Python, Pytest, Shell Scripting, Jenkins, Git,
Testing: Manual & Automation Testing, SDLC, STLC, Agile Methodology, Bug Life Cycle, Bug Tracking
Servers & Virtualization: BIOS, BMC, CPLD, VM's, Snapshots, Cloning, HA, FT, DRS
Operating Systems: RHEL 9.4, Windows (2019, 2022, 2025), VMWare ESXI (7.0, 8.0, 9.0)

PROFESSIONAL EXPERIENCE

Company Name – Sep 2023 - present
Role: Storage QA Engineer

PROJECTS

Enterprise Storage Platform & Feature Validation – DDN IntelliFlash

Client: DDN (DataDirect Networks) **(**Sep 2023 – Aug 2025)

Project Description

Worked on the validation of the DDN IntelliFlash enterprise all-flash storage platform, ensuring hardware, firmware, and software quality. Performed platform and feature validation across multiple storage configurations and firmware versions. Executed functional, regression, compatibility, and performance testing to verify reliability and stability.

Roles & Responsibilities

- Validated enterprise storage components including controllers, BIOS, BMC, CPLD, SSD/HDD/NVMe, HBAs, PCIe, networking, power supplies, and enclosures across hardware/firmware configurations.
- Validated BIOS/BMC functionality including boot, device enumeration, sensors, event logs, remote console, power control, firmware updates, reset, and recovery.
- Executed platform fault/recovery scenarios covering controller/BMC/BIOS resets, reboots, power cycles, firmware failures, and device recovery.
- Performed RAID, storage pool, volume/LUN, hot-spare, degraded-mode, rebuild, recovery, compatibility, firmware, and fault-injection validation.
- Validated FC/iSCSI connectivity, multipathing, path/controller failover-failback, and continuous I/O during reboot and power-cycle scenarios.
- Conducted FIO-based performance, stress, and endurance testing, analyzing IOPS, throughput, latency, block sizes, queue depths, and workloads.
- Developed Python/Pytest automation for platform health, BIOS/BMC validation, storage monitoring, log collection, and regression, reducing manual effort by 50%.
- Analyzed platform, controller, drive, OS, BMC, and performance logs to identify failures, error patterns, and root causes.
- Managed VMware ESXi/vCenter environments including VM provisioning, snapshots, cloning, templates, and virtual infrastructure administration.
- Leveraged ChatGPT/OpenAI and GitHub Copilot for automation development, test generation, code optimization, troubleshooting, and AI-assisted log analysis.

Enterprise HDD/SSD/NVMe Drive Qualification – DDN A3I

Client: DDN (DataDirect Networks) Aug 2025 – present

Project Description

Performed enterprise drive validation for the DDN A3I storage platform, focusing on HDD, SSD, and NVMe qualification, compatibility, performance, reliability, firmware, and fault-recovery testing. Developed Python-based automation and leveraged AI-assisted tools for test development, log analysis, troubleshooting, and validation efficiency.

Roles & Responsibilities

- Validated SSD features including SMART health monitoring, wear leveling, garbage collection, TRIM, over-provisioning, endurance, secure erase, and firmware compatibility to ensure drive reliability, performance, and data integrity.
- Performed NVMe feature validation including namespace management, SMART log analysis, firmware downloads and activation, power state transitions, PCIe link validation, error recovery, and hot-plug functionality across enterprise storage platforms.
- Verified drive detection, capacity, firmware, interface speed, health status, compatibility, and behavior across supported hardware configurations.
- Conducted performance benchmarking using FIO, analyzing IOPS, throughput, latency, bandwidth, and mixed workloads to validate drive performance.
- Performed stress, endurance, power-cycle, hot-plug, fault-injection, and long-duration I/O testing to ensure drive reliability and stability.
- Validated RAID functionality during drive failures, degraded states, hot-spare activation, rebuild operations, and recovery scenarios.
- Executed drive firmware upgrade and downgrade validation, ensuring compatibility, stability, and data integrity.
- Diagnosed storage issues using smartctl, nvme-cli, storcli, dmesg, and Linux system logs to perform root-cause analysis.
- Developed Python/Pytest automation for drive discovery, health monitoring, FIO execution, log collection, and regression validation, reducing manual regression effort by 60%.
- Leveraged Claude and GitHub Copilot for automation development, test-scenario generation, log-pattern analysis, debugging, and optimization of storage validation workflows.
- Documented defects in JIRA and collaborated with development and firmware teams for root-cause analysis and fix verification.

EDUCATION

Bachelor of Science (Computer Science) | PRGC(A)**.** **2020 – 2023**